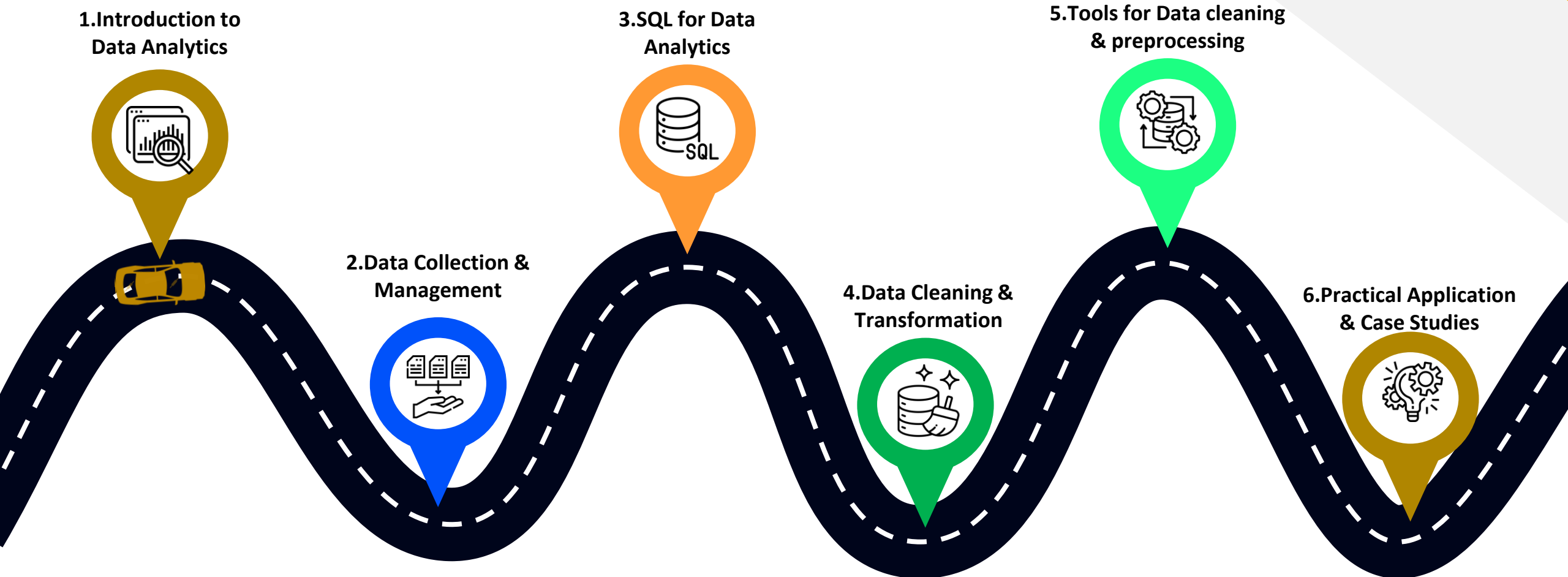




Part One: Introduction

Course Road Map



Overview Of Data Analysis

- Data Analysis Definition
- Data Analysis VS Data Analytics
- Real World Business Cases
- Data Analysis Applications



What is Data Analysis

- **Data analysis** is the practice of examining data to answer questions, identify trends, and extract insights.
- When data analytics is used in business, it's often called Business Analytics



Data Analysis VS Data Analytics

Data Analysis

- Focuses on examining and interpreting historical data to identify patterns and insights.
- Aims to understand what has happened by summarizing and visualizing data.
- Utilizes basic tools like Excel and SQL for querying and visualizing data.

Data Analytics

- Encompasses data analysis but also includes predictive and prescriptive techniques to inform decisions.
- Aims to predict what will happen and prescribe actions based on data
- Involves advanced tools and techniques, including statistical software and machine learning libraries.

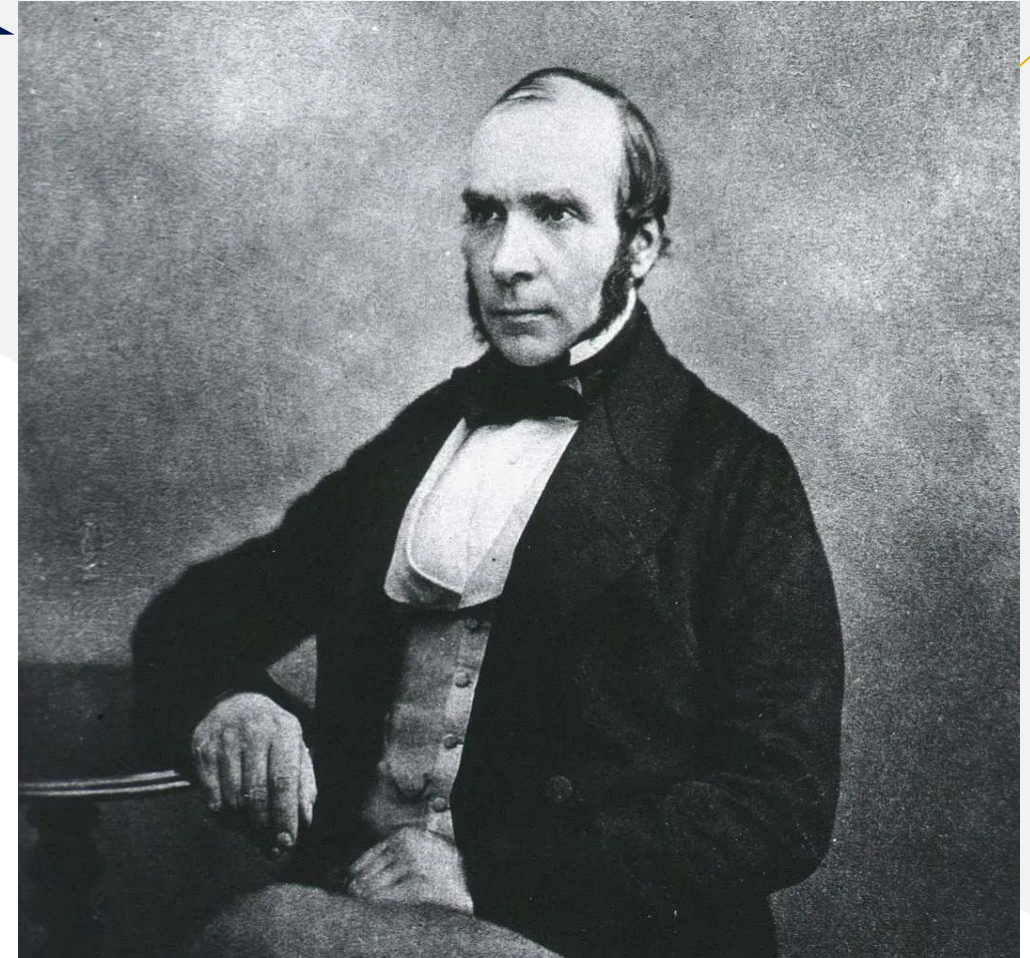
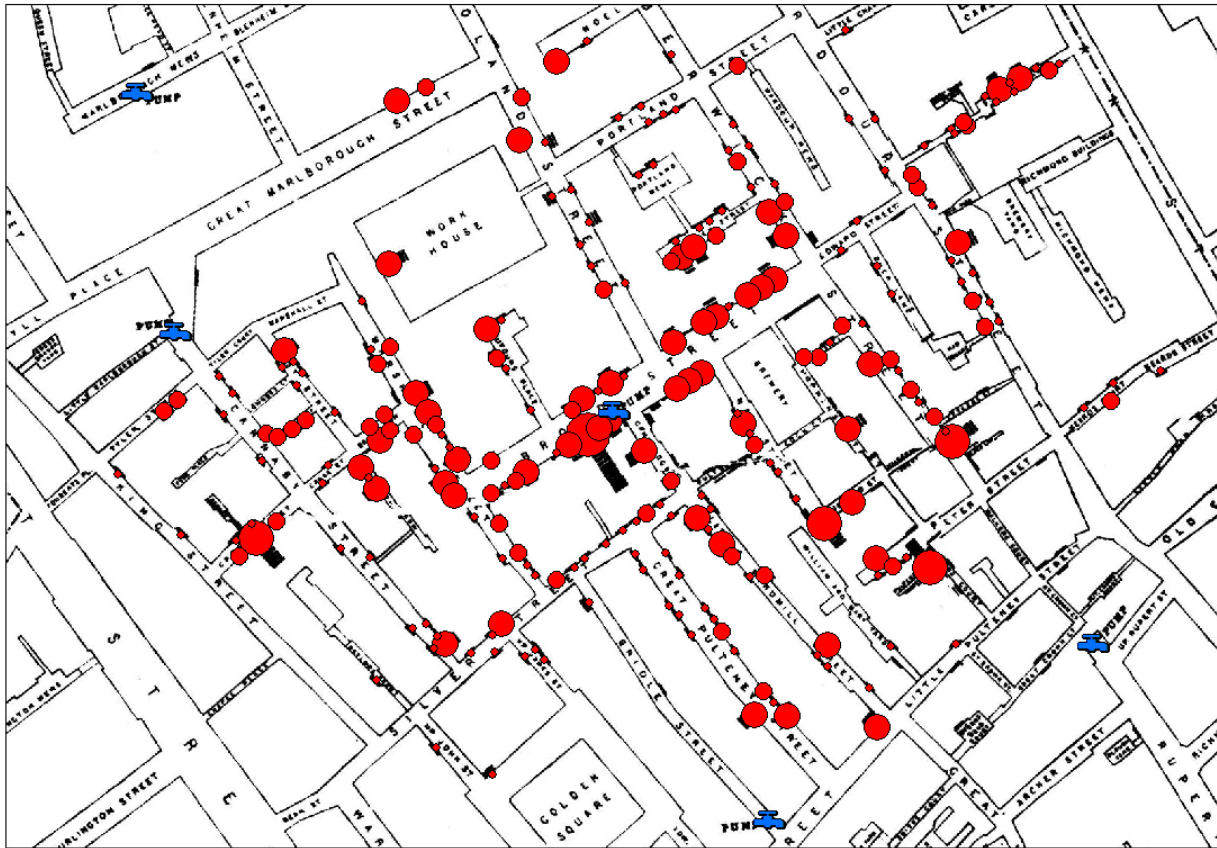
Data Analysis VS Data Analytics

	Data Analysis	Data Analytics
Scope	Current Being	Current Being & Future
Goals	Describe & Diagnose	Predict & Prescript
Tools	Excel – SQL	Statistical & ML tools

Data Analytics Applications

- 1.Retail & E-Commerce
- 2.Healthcare
- 3.Finance & Banking
- 4.Manufacturing & Supply Chain
- 5.Marketing & Social Media





Types Of Data Analytics

- **Descriptive Analytics**
- **Diagnostic Analytics**
- **Predictive Analytics**
- **Prescriptive Analytics**



Descriptive Data Analytics

- Answers the question, "What happened?"
by summarizing historical data to identify patterns and trends.
- Uses SQL, Excel, Power BI, ...
- Performed by Data Analysts or Business Analysts



Diagnostic Data Analytics

- Answers the question, "Why did it happen?" by drilling down into data to discover the causes of past outcomes.
- Uses Python, R, SQL, ...
- Performed by Data Analysts and Business Analysts



Predictive Data Analytics

- Answers the question, "What is likely to happen?" by using statistical models and machine learning techniques to forecast future outcomes based on historical data.
- Uses Python, R, SAS, IBM SPSS
- Performed by Data Scientist and Machine Learning Engineers



Prescriptive Data Analytics :

- Answers the question, "What should we do?" by recommending actions to achieve desired outcomes using optimization and simulation techniques.
- Uses Python, R, SAS, IBM ILOG CPLEX Optimization Studio.
- Performed by Data Scientist and Operations Research Analyst.



TYPES OF DATA ANALYTICS



Questions

- What were our total sales last month?
- Why did sales drop at the Airport store in February?
- Will our sales improve next month if the weather stays sunny?
- What marketing strategy should we use next month to boost sales?
- Which store had the highest number of promo code redemptions?

Data Analytics Process

- **Set SMART Business Goals**
- **Determine KPIs**
- **OSEMN Data Analytics Process**
- **(Bonus) Job Profiles**
- **Case Study**



SMART Business Goals



SPECIFIC

- State what you'll do
- Use action words



MEASURABLE

- Provide a way to evaluate
- Use metrics or data targets



ACHIEVABLE

- Within your scope
- Possible to accomplish, attainable



RELEVANT

- Makes sense within your job function
- Improves the business some way



TIME-BOUND

- State when you'll get it done
- Be specific on date or timeframe

Determine KPIs

- **KPIs are measurable values that can help you track your progress towards your goal.**
- **Usually quantitative**
- **Align with and reflect strategic goals.**
- **Must be clear, specific, and quantifiable.**
- **Provide insights for decision-making and improvements.**



OSEMN Framework

Data Science Process



Obtain



Scrub



Explore



Model



iNterpret

Case Study



Bean & Brew Coffee Shop Sales

Data Analytics Platforms And Environments

- **Excel For Data Analysis**
- **Power BI & Power Services**
- **Tableau For Data Visualization**
- **Python For Data Analysis**
- **R For Data Analysis**



Excel



SQL
(Structured
Query Language)



Power BI



Tableau



Python



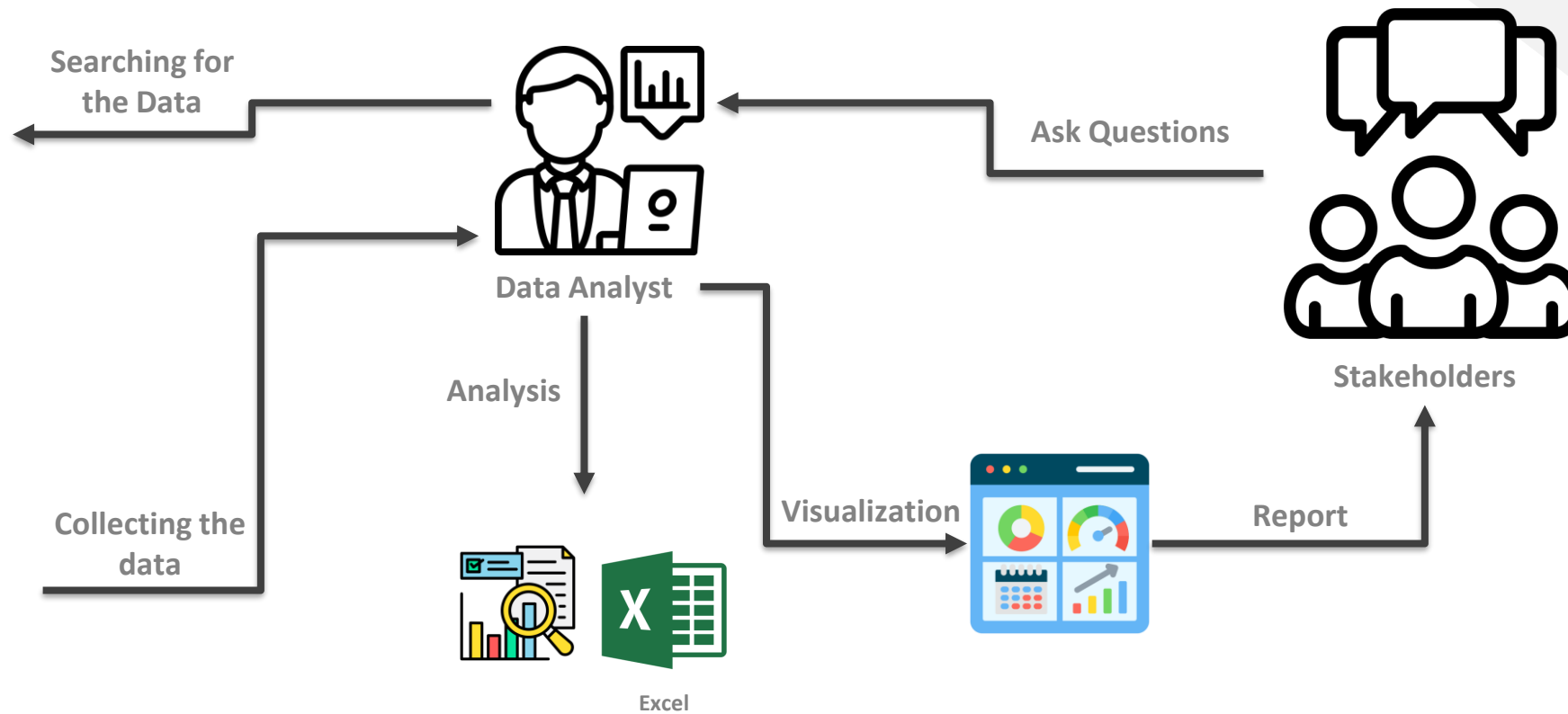
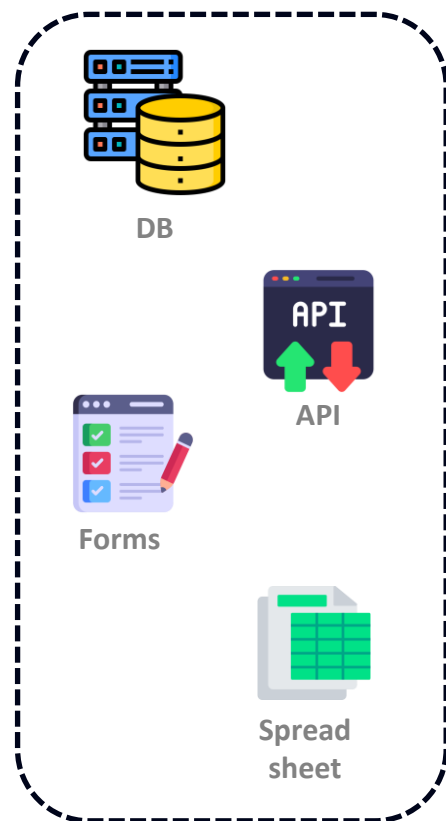
R

Data Analytics Job Profiles



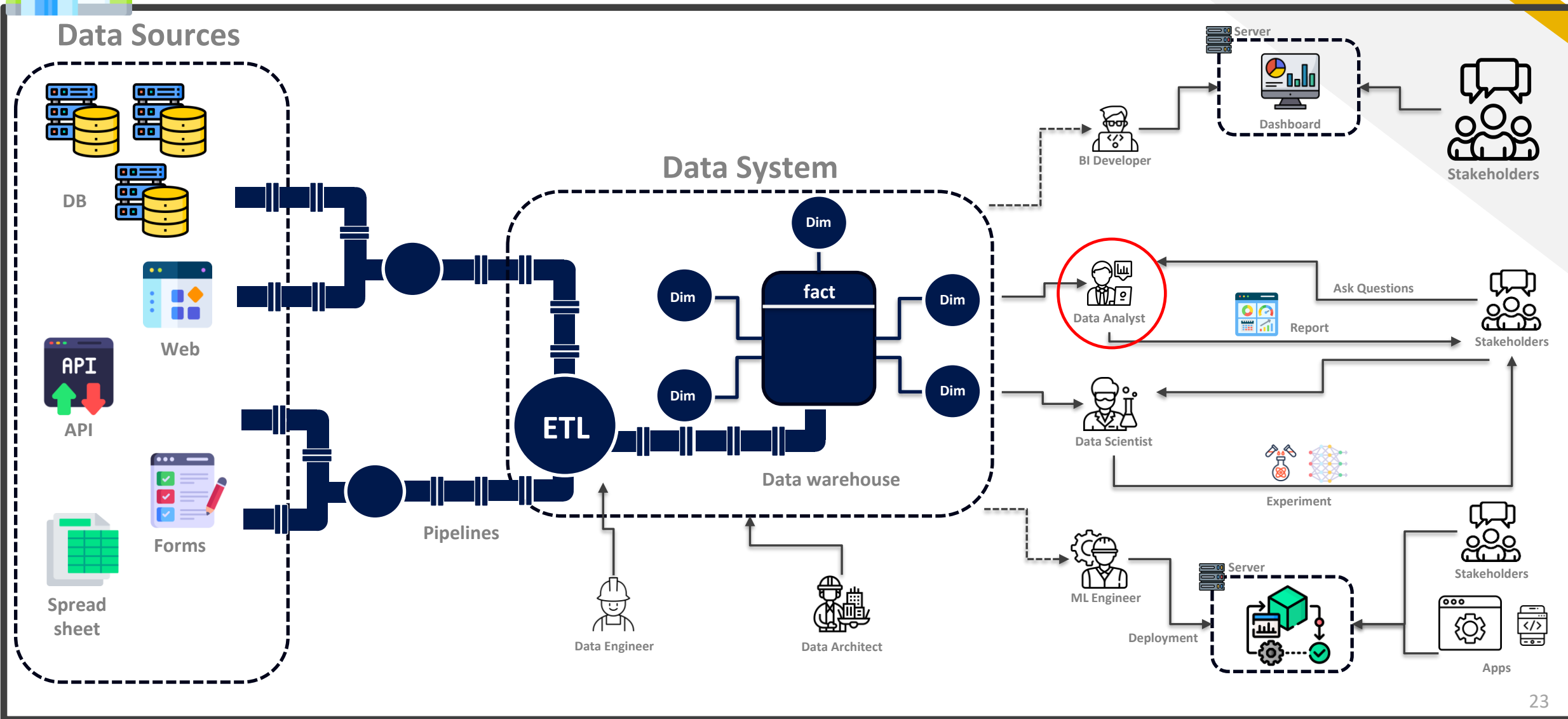
Small Company

Data Sources



Data Analytics Job Profiles

Big Company



Questions





Thank You.



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